

Basic CGE Model Series – Algebraic Introduction

Xilong Yan

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0 Overview

Basic CGE model series offer a clean, canonical framework that helps us better understand the theoretical basis of general equilibrium models and provide new perspectives.

A prominent feature of this series is the formulation – we express the models in the form of non-linear inequalities system, or the so-called Mixed Complementarity Problem (MCP), rather than traditional equalities system. With this new expression approach, the model equations are comprised of three parts:

1. **Zero-profit conditions**, which determine ‘quantities’;
2. **Market clearance conditions**, which determine ‘prices’;
3. **Other equations**, including e.g., calculation of intermediate variables and indicators.

We are going to build many model versions, starting from the easiest and becoming more and more complicated.

1 BA-CGE v1.0

This is a simple **single-region, single-sector, two-factor, close-economy** CGE model. The firm maximizes its profit subjected to constant return-to-scale CES technology. The households maximize its utility (following a log utility function) subjected to the budget constraint. There are no governments, investments, nor the exports/imports. In total, there are three markets: the goods market and two factor markets.

1.1 Structure

See Figure 1.1.

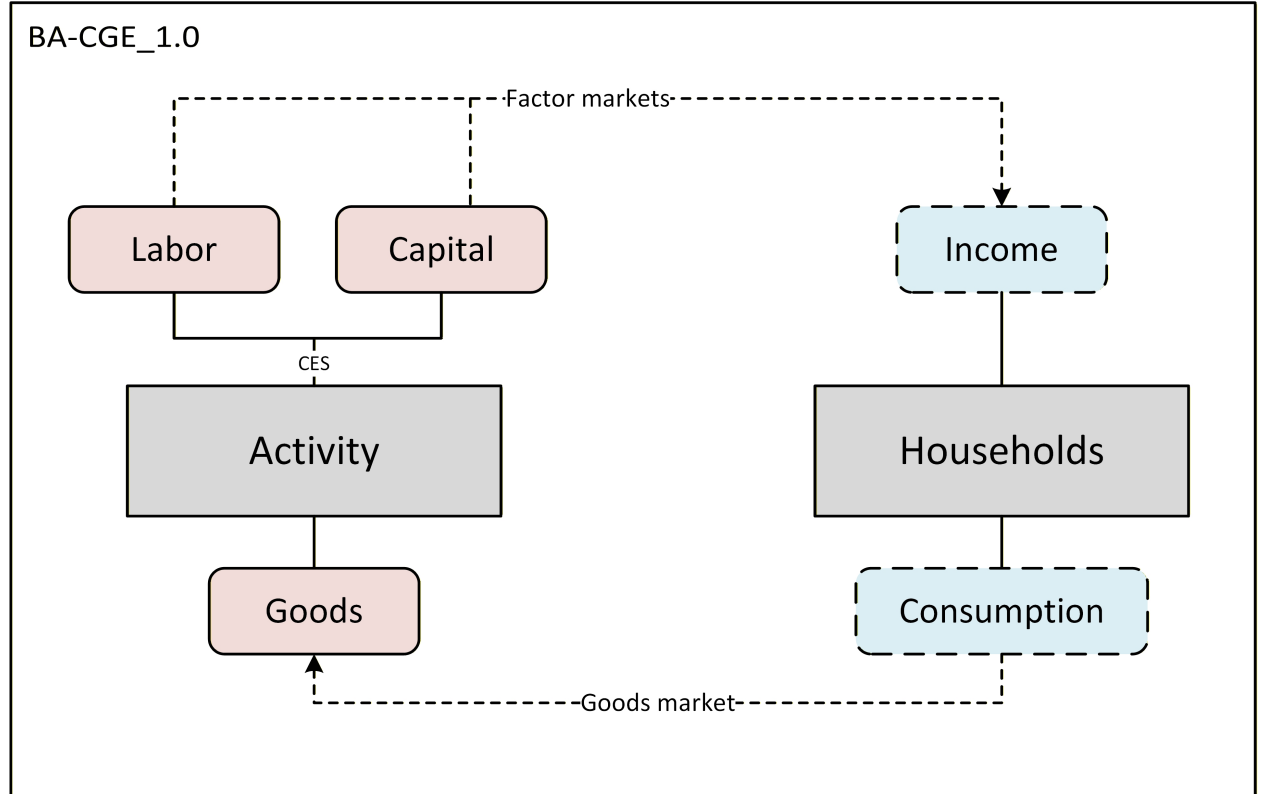


Figure 1.1: Model structure of BA-CGE v1.0

1.2 Notations

See Tables 1.1 – 1.3.

1.3 Model equations

Production module:

- Intermediate calculation – Conditional labor demand function (L_{eq}):

$$L = \frac{\partial \Pi}{\partial W} Q = \lambda_l^{\sigma-1} \alpha_l \left(\frac{P}{W} \right)^\sigma Q \quad (1.1)$$

- Intermediate calculation – Conditional capital demand function (K_{eq}):

$$K = \frac{\partial \Pi}{\partial R} Q = \lambda_k^{\sigma-1} \alpha_k \left(\frac{P}{R} \right)^\sigma Q \quad (1.2)$$

Table 1.1: Model variables

Variables	Descriptions
L	Quantities of labor
K	Quantities of capital
Q	Quantities of goods
U	Values of households utility
Y	Values of households income
W	Price of labor (wages)
R	Price of capital (rents)
P	Price of goods
D	Dummy variable for Walras test

Table 1.2: Model parameters

Parameters	Descriptions
σ	Substitution elasticity between labor and capital
α_l	Cost share of labor in production function
α_k	Cost share of capital in production function
a	Total factor productivity in production function
λ_l	Labor-augmented technical shifter
λ_k	Capital-augmented technical shifter
$\bar{L}$	Exogenous labor endowment
$\bar{K}$	Exogenous capital endowment

Table 1.3: Other notations

Parameters	Descriptions
Π	Unit profit of production (the net difference between unit revenue and unit cost)

- Zero-profit condition – Goods output (Q_{eq}):

$$0 \geq \Pi = \begin{cases} P - \left[\alpha_l \left(\frac{W}{\lambda_l} \right)^{1-\sigma} + \alpha_k \left(\frac{R}{\lambda_k} \right)^{1-\sigma} \right]^{\frac{1}{1-\sigma}}, & \sigma \neq 1 \\ P - \frac{1}{a} \left[\left(\frac{1}{\lambda_l} \frac{W}{\alpha_l} \right)^{\alpha_l} \left(\frac{1}{\lambda_k} \frac{R}{\alpha_k} \right)^{\alpha_k} \right], & \sigma = 1 \end{cases} \quad \perp \quad Q \quad (1.3)$$

Households module:

- Indicator calculation – Households utility function (U_{eq}):

$$U = \ln Q \quad (1.4)$$

- Intermediate calculation – Households income balance (Y_{eq}):

$$Y = WL + RK \quad (1.5)$$

Market clearance:

- Market clearance – Labor (W_{eq}):

$$0 \geq L - \bar{L} \quad \perp \quad W \quad (1.6)$$

- Market clearance – Capital (R_{eq}):

$$0 \geq K - \bar{K} \quad \perp \quad R \quad (1.7)$$

- Market clearance – Goods (P_{eq})¹:

$$D = \frac{Y}{P} - Q \quad (1.8)$$

1.4 Extensions to v1.0

BA-CGE v1.1 adds households savings and investments into the model, and v1.2 adds government into the model. For these versions, refer to *Model update log* document.

¹Note that this equation is dummied because P is set as the numeraire.

2 BA-CGE v2.0

This is a simple **single-region, single-sector, two-factor, open-economy** CGE model. The firm maximizes its profit subjected to constant return-to-scale CES technology. The households save a fixed proportion of its income into the bank, and then maximize its utility (increasing and convex) subjected to the budget constraint. The government levies a production tax, a direct tax and an import tariff, and uses its income to purchase. It operates with a constant fiscal surplus by lending to/borrowing from the bank. The investment bank absorbs households savings and government net surplus, and forms into investment. Imports are specified using Armington assumption, while exports follow the CET function. In total, there are seven markets: five goods markets (activity output, exports, domestic goods, imports, composite goods) and two factor markets.

2.1 Structure

See Figure 2.1.

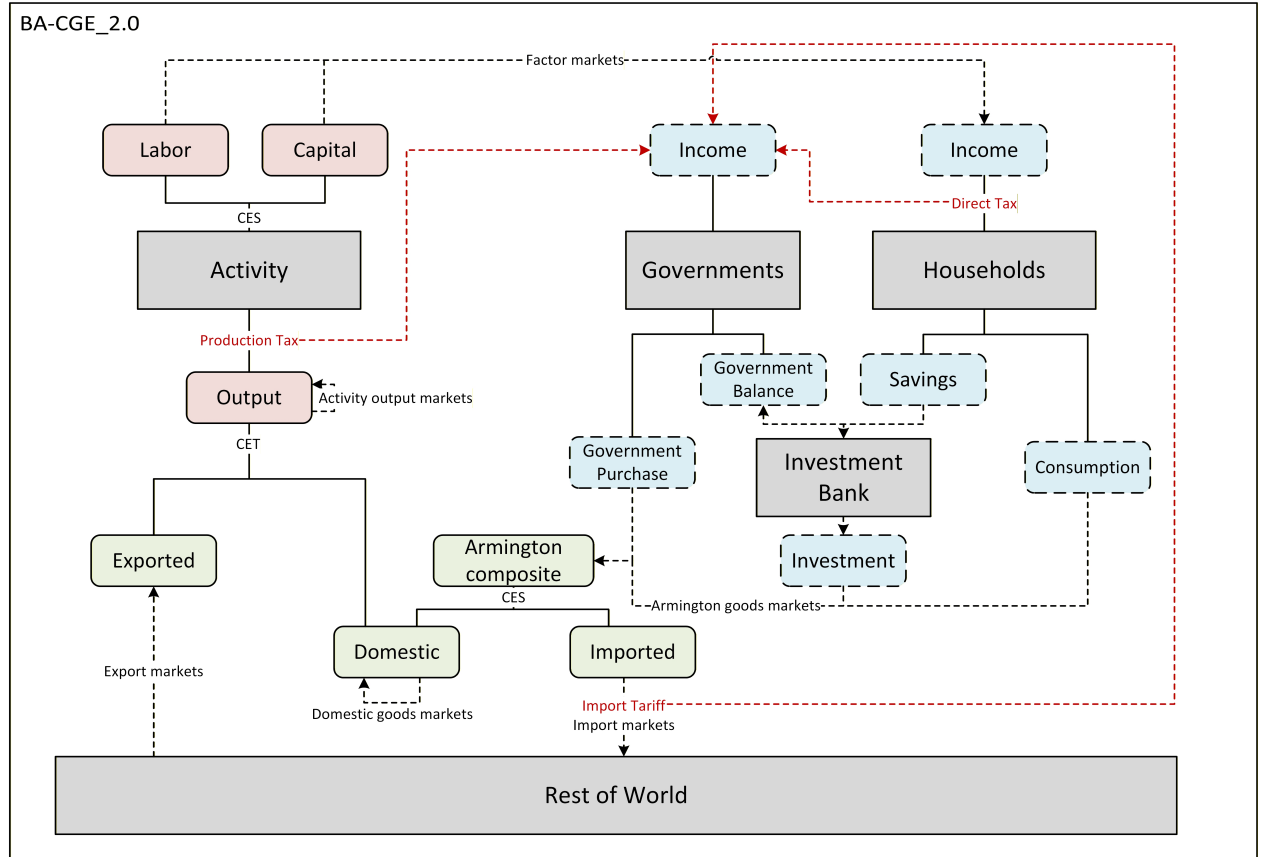


Figure 2.1: Model structure of BA-CGE v2.0

2.2 Notations

See Tables 2.1 – 2.7.

2.3 Model equations

Production module:

Table 2.1: Model variables – Prices

Variables	Descriptions
P^{At}	Price of activity output (pre-tax)
P^{Mt}	Price of imported goods (pre-tax)
P^P	Price of households composite goods
P^G	Price of government composite goods
P^I	Price of investment composite goods
P^L	Price of labor (wages)
P^K	Price of capital (rents)
P^A	Price of activity output
P^D	Price of domestic goods
P^E	Price of exported goods
P^M	Price of imported goods
P^C	Price of composite goods

Table 2.2: Model variables – Quantities

Variables	Descriptions
Q^L	Quantities of labor demand
Q^K	Quantities of capital demand
Q^A	Quantities of activity output supply
Q^D	Quantities of domestic goods supply
Q^E	Quantities of exported goods supply
Q^M	Quantities of imported goods demand
Q^C	Quantities of composite goods supply
Q^P	Quantities of households composite goods
Q^G	Quantities of government composite goods
Q^I	Quantities of investment composite goods

Table 2.3: Model variables – Values and others

Variables	Descriptions
S^F	Values of net foreign savings
C^P	Values of households consumption
S^P	Values of households savings
Y^P	Values of households income
C^G	Values of government purchase
S^G	Values of government fiscal surplus
Y^G	Values of government income
Y^I	Values of investment expenditures
D	Dummy variable for Walras test

Table 2.4: Model parameters – Elasticities

Parameters	Descriptions
σ^t	Substitution elasticity between labor and capital
ω^x	Transformation elasticity between domestic and exported goods
σ^m	Substitution elasticity between domestic and imported goods

Table 2.5: Model parameters – Structures

Parameters	Descriptions
α_l	Cost share of labor in production function
α_k	Cost share of capital in production function
a_a	Scale parameter in production function
λ_l	Labor-augmented technical shifter
λ_k	Capital-augmented technical shifter
γ_d	Cost share of domestic goods in export CET function
γ_e	Cost share of exported goods in export CET function
α_d	Cost share of domestic goods in armington import nest function
α_m	Cost share of imported goods in armington import nest function
a_m	Scale parameter in armington import nest function
τ^p	Tax rate on production
τ^m	Tax rate on imports
β^s	Households saving propensity
τ^h	Tax rate on households income

Table 2.6: Model parameters – Closures

Parameters	Descriptions
$\bar{S}^G$	Exogenous real government savings
$\bar{Q}^L$	Exogenous labor supply
$\bar{Q}^K$	Exogenous capital supply
$\bar{Q}^E$	Exogenous export demand
$\bar{p}^{Ew}$	Fixed world price of exported goods
$\bar{p}^{Mw}$	Fixed world price of imported goods

Table 2.7: Other notations

Parameters	Descriptions
Π^A	Unit profit of production (the net difference between unit revenue and unit cost)
Π^E	Unit profit of export CET
Π^C	Unit profit of Armington composite
Π^P	Unit profit of households composite
Π^G	Unit profit of government composite
Π^I	Unit profit of investment composite
GDP	Gross domestic production

- Intermediate calculation – Conditional demand function of labor (QL_eq):

$$Q^L = \frac{\partial \Pi^A}{\partial P^L} Q^A = \lambda_l^{\sigma^t - 1} \alpha_l \left(\frac{P^{At}}{P^L} \right)^{\sigma^t} Q^A \quad (2.1)$$

- Intermediate calculation – Conditional demand function of capital (QK_eq):

$$Q^K = \frac{\partial \Pi^A}{\partial P^K} Q^A = \lambda_k^{\sigma^t - 1} \alpha_k \left(\frac{P^{At}}{P^K} \right)^{\sigma^t} Q^A \quad (2.2)$$

- Zero-profit condition – Activity output (supply) (QA_eq):

$$0 \geq \Pi^A = \begin{cases} P^{At} - \left[\alpha_l \left(\frac{P^L}{\lambda_l} \right)^{1-\sigma^t} + \alpha_k \left(\frac{P^K}{\lambda_k} \right)^{1-\sigma^t} \right]^{\frac{1}{1-\sigma^t}}, & \sigma^t \neq 1 \\ P^{At} - \frac{1}{a_a} \left[\left(\frac{1}{\lambda_l} \frac{P^L}{\alpha_l} \right)^{\alpha_l} \left(\frac{1}{\lambda_k} \frac{P^K}{\alpha_k} \right)^{\alpha_k} \right], & \sigma^t = 1 \end{cases} \perp Q^A \quad (2.3)$$

- Intermediate calculation – Calculation of pre-tax price (production tax) (PAT_eq):

$$P^{At} = \frac{P^A}{1 + \tau^p} \quad (2.4)$$

Trade module:

- Intermediate calculation – Conditional supply function of domestic goods (QD_eq)²:

$$\begin{cases} Q^D = \frac{\partial \Pi^E}{\partial P^D} Q^A = \gamma_d \left(\frac{P^D}{P^A} \right)^{\omega^x} Q^A, & \omega^x \neq \infty \\ P^D = P^A, & \omega^x = \infty \end{cases} \quad (2.5)$$

- Intermediate calculation – Conditional supply function of exported goods (QE_eq):

$$\begin{cases} Q^E = \frac{\partial \Pi^E}{\partial P^E} Q^A = \gamma_e \left(\frac{P^E}{P^A} \right)^{\omega^x} Q^A, & \omega^x \neq \infty \\ P^E = P^A, & \omega^x = \infty \end{cases} \quad (2.6)$$

- Zero-profit condition – Activity output (demand) (QA_eq)³:

$$0 \geq \begin{cases} \Pi^E = \left(\gamma_d P^{D1+\omega^x} + \gamma_e P^{E1+\omega^x} \right)^{\frac{1}{1+\omega^x}} - P^A, & \omega^x \neq \infty \\ (Q^D + Q^E) - Q^A, & \omega^x = \infty \end{cases} \perp Q^A \quad (2.7)$$

- Intermediate calculation – Calculation of pre-tax price (import tariff) (PMT_eq):

$$P^{Mt} = \frac{P^M}{1 + \tau^m} \quad (2.8)$$

²Note that when $\omega^x = \infty$, the supply of domestic goods is governed by law-of-one-price. Similar expressions apply here-in-after.

³Note that this equation is omitted and is essentially substituted by the market clearance PA_eq .

- Intermediate calculation – Conditional demand function of domestic goods (QD_eq)⁴:

$$Q^D = \frac{\partial \Pi^C}{\partial P^D} Q^C = \alpha_d \left(\frac{P^C}{P^D} \right)^{\sigma^m} Q^C \quad (2.9)$$

- Intermediate calculation – Conditional demand function of imported goods (QM_eq):

$$Q^M = \frac{\partial \Pi^C}{\partial P^M} Q^C = \alpha_m \left(\frac{P^C}{P^M} \right)^{\sigma^m} Q^C \quad (2.10)$$

- Zero-profit condition – Composite goods (QC_eq):

$$0 \geq \Pi^C = \begin{cases} P^C - \left(\alpha_d P^{D^{1-\sigma^m}} + \alpha_m P^{M^{1-\sigma^m}} \right)^{\frac{1}{1-\sigma^m}}, & \sigma^m \neq 1 \\ P^C - \frac{1}{\alpha_m} \left[\left(\frac{P^D}{\alpha_d} \right)^{\alpha_d} \left(\frac{P^M}{\alpha_m} \right)^{\alpha_m} \right], & \sigma^m = 1 \end{cases} \perp Q^C \quad (2.11)$$

- Intermediate calculation – Calculation of net foreign savings (YFS_eq):

$$S^F = P^{Mt} Q^M - P^E Q^E \quad (2.12)$$

Households module:

- Zero-profit condition – Households composite goods (QP_eq):

$$0 \geq \Pi^P = P^P - P^C \perp Q^P \quad (2.13)$$

- Market clearance – Households composite goods (PP_eq):

$$0 \geq C^P - P^P Q^P \perp P^P \quad (2.14)$$

- Intermediate calculation – Calculation of households consumption (YPC_eq):

$$C^P = (1 - \beta^s) Y^P \quad (2.15)$$

- Intermediate calculation – Calculation of households savings (YPS_eq):

$$S^P = \beta^s Y^P \quad (2.16)$$

- Intermediate calculation – Households income balance (YP_eq):

$$Y^P = (1 - \tau^h) (P^L Q^L + P^K Q^K) \quad (2.17)$$

Government module:

- Zero-profit condition – Government composite goods (QG_eq):

$$0 \geq \Pi^G = P^G - P^C \perp Q^G \quad (2.18)$$

- Market clearance – Government composite goods (PG_eq):

$$0 \geq C^G - P^G Q^G \perp P^G \quad (2.19)$$

⁴Note that this equation is omitted and is essentially substituted by the market clearance PD_eq.

- Intermediate calculation – Calculation of government consumption (YGC_eq):

$$C^G = Y^G - S^G \quad (2.20)$$

- Intermediate calculation – Calculation of government savings (YGS_eq):

$$S^G = \bar{S}^G P^G \quad (2.21)$$

- Intermediate calculation – Government income balance (YG_eq):

$$Y^G = \tau^p Q^A P^{At} + \tau^m Q^M P^{Mt} + \tau^h (P^L Q^L + P^K Q^K) \quad (2.22)$$

Investment module:

- Zero-profit condition – Investment composite goods (QI_eq):

$$0 \geq \Pi^I = P^I - P^C \perp Q^I \quad (2.23)$$

- Market clearance – Investment composite goods (PI_eq):

$$0 \geq Y^I - P^I Q^I \perp P^I \quad (2.24)$$

- Intermediate calculation – Savings-investment balance (YI_eq):

$$Y^I = S^P + S^G + S^F \quad (2.25)$$

Market Clearance:

- Market clearance – Labor (PL_eq):

$$0 \geq Q^L - \bar{Q}^L \perp P^L \quad (2.26)$$

- Market clearance – Capital (PK_eq):

$$0 \geq Q^K - \bar{Q}^K \perp P^K \quad (2.27)$$

- Market clearance – Activity output (PA_eq):

$$0 \geq Q^A(d) - Q^A(s) \perp P^A \quad (2.28)$$

- Market clearance – Domestic goods (PD_eq):

$$0 \geq Q^D(d) - Q^D(s) \perp P^D \quad (2.29)$$

- Market clearance – Exported goods (PE_eq):

$$0 = \begin{cases} P^E - \bar{p}^{Ew} P^C \\ Q^E - \bar{Q}^E \end{cases} \perp P^E \quad (2.30)$$

- Market clearance – Imported goods (PM_eq):

$$0 = P^{Mt} - \bar{p}^{Mw} P^C \perp P^M \quad (2.31)$$

- Market clearance – Composite goods (PC_eq)⁵:

$$D = (Q^P + Q^G + Q^I) - Q^C \quad (2.32)$$

Indices:

- Indicator calculation – GDP (GDP_eq):

$$GDP = P^P Q^P + P^G Q^G + P^I Q^I + P^E Q^E - P^{Mt} Q^M \quad (2.33)$$

⁵Note that this equation is dummied because PC is set as the numeraire.